Lie Groups — Notes

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1 Journal 1

We will provide another definition of a smooth manifold and show equivalency to the atlas based definition.

Definition 1.1 (Locally diffeomorphic smooth manifold)**.** $X \subseteq \mathbb{R}^n$ is called a k-manifold if for any $p \in X$ there is an open neighborhood $p \in W \subseteq X$ (open in the subspace topology induced by X) and an open set $V \subseteq \mathbb{R}^k$, and a smooth invertible map $\varphi : V \to W$, namely a diffeomorphism.
In other words X is a manifold if it is locally diffeomorphic to open sets in $\mathbb{R}^k$.

In order for our map φ to actually be considered a diffeomorphism we have to define differentiability of its invertible map $\psi : W \to V$. To do so, we define differentiability of such map to be such that when extending W to an open set $W \subseteq W' \subseteq \mathbb{R}^n$, there is a function $\Phi : W' \to \mathbb{R}^m$ such that $\Phi \mid_{W \cap W'} = \psi$.

Theorem 1.2 (Equivalency of manifold definitions)**.** *$X \subseteq \mathbb{R}^n$ is a smooth manifold iff it is a locally diffeomorphic manifold.*

Proof. $1 \implies 2$: Let $A = \{\varphi : U_\alpha \to X_\alpha \mid \alpha \in I\}$ be an atlas of $X \subseteq \mathbb{R}^n$ and let $p \in X$ be some point. $\bigcup_{\alpha \in I} X_\alpha = X$ by the atlas definition, therefore there is $\alpha \in I$ such that $p \in X_\alpha$, therefore φ_α^{-1} is a local diffeomorphism fulfilling the local diffeomorphism property.

$2 \implies 1$: Let X be a locally diffeomorphic manifold and let $I = X$. For each $\alpha \in I$ let $\varphi_\alpha^{-1} : W_\alpha \to X_\alpha$ be a local diffeomorphism of $\alpha \in X$. Define $A = \{\varphi_\alpha \mid \alpha \in I\}$, this is an atlas as $\alpha \in X_\alpha$ for all α so $\bigcup_{\alpha \in I} X_\alpha = X$. □

Notice that we did not check for Hausdorff nor for 2nd countability, as the euclidean spaces (equipped with the standard topology) are both.

Remark. The definition of a manifold induced by an atlas discards the requirement to take $X \subseteq \mathbb{R}^n$, instead letting us discuss manifolds as a general property.
Even by doing so, it is easy to emphasize the concept of manifolds in euclidean spaces.